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Sri Sivasubramaniya Nadar College of Engineering, Kalavakkam – 603 110
(An Autonomous Institution, Affiliated to Anna University, Chennai)

Computer Science and Engineering
Continuous Assessment Test – I Answer Key
Regulations – R2021

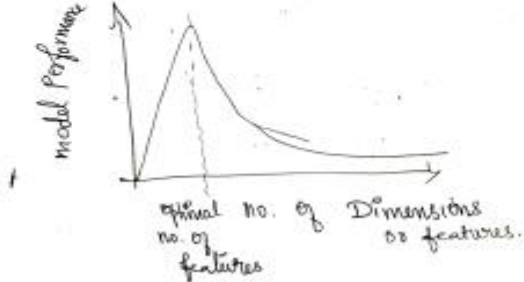
Degree & Branch	BE - CSE				Semester	6
Subject Code & Name	UCS2604 - PRINCIPLES OF MACHINE LEARNING					
Academic Year	2023-2024 ODD/EVEN	Batch	2021-2025	Date	18.03.2024	FN /AN
Time: 08:10 - 09:40 a.m (90 Minutes)	Answer All Questions				Maximum: 50 Marks	

(K1: Remembering, K2: Understanding, K3: Applying, K4: Analyzing, K5: Evaluating)

CO1	Explain the basic concepts of machine learning (K2)
CO2	Apply supervised algorithms for different classification problems (K3)
CO3	Explain the need for ensemble methods (K2)
CO4	Apply unsupervised and reinforcement learning techniques to various problems (K3)
CO5	Apply dimensionality reduction and optimization techniques (K3)

Part – A (4 × 2 = 8 Marks)

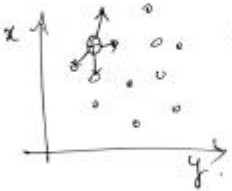
		KL	CO	PI
1.	Compare Classification and regression problems with suitable examples. <i>Classification : Finding out the category of the input test data (Class label of through trained ML model / Test data)</i> <i>Eg: Given patient information Identify whether patient has a disease or not.</i> <i>Discrete output</i> <i>Regression : Finding out the single output value for the given input test data</i> <i>Applied for continuous values of input</i> <i>Eg: Identify the price of stock.</i>	K2	CO1	
2.	Summarize Curse of Dimensionality with suitable graph	K2	CO1	

	Curse of Dimensionality ③ Many dimensions in the input dataset would mislead the model due to irrelevant features & redundant features too. 			
3.	Consider the company's performance over a five-year period, the closing price of stocks for a five year period is given below. • \$74.01 • \$74.77 • \$73.94 • \$73.61 • \$73.40 Apply the method of sum of squares and infer whether the customers could invest in the company's stocks or not. Ans: The sum of the total prices is \$369.73 and the mean or average price is $\\$369.73 \div 5 = \\73.95. Then, figure out the sum of squares, we find the difference of each price from the average, square the differences, and add them together: • sum of squares = $(\\$74.01 - \\$73.95)^2 + (\\$74.77 - \\$73.95)^2 + (\\$73.94 - \\$73.95)^2 + (\\$73.61 - \\$73.95)^2 + (\\$73.40 - \\$73.95)^2$ • sum of squares = $(0.06)^2 + (0.82)^2 + (-0.01)^2 + (-0.34)^2 + (-0.55)^2$ • sum of squares = 1.0942 Investor can still invest in the stocks as the sum of squares value is positive.	K3	CO2	
4.	Interpret the need for kernel trick in Support Vector Machines ④ Kernel trick would try to classify non-linear data by mapping the dimensions into the higher dimensional space. $f(x) \xrightarrow{\text{Transform to}} \phi(x)$	K2	CO2	

Part – B (3×6 = 18 Marks)

	KL	CO	PI
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5.	In the simplest case, a classification consists of two possible outcomes. Suppose we want to investigate how well Corona tests reflect the infection status of a patient. In this case, the Corona test serves as a classifier of a total of two outcomes: infected or non-infected. Given a sample of 13 individuals which includes both infected and non-infected individuals. The dataset has 8 infected and 5 non-infected individuals, where infected belong to class 1 and non-infected belong to class 0. The actual and predicted class labels are shown below. Actual Class = [1,1,1,0,0,0,0,1,0,0,0,0,0] Predicted Class = [0,0,0,1,1,1,1,1,0,0,0,0,0] Construct a confusion matrix and calculate precision, recall and F1 measure. <u>Part-B</u> 5. Total = 13. 8 - infected, 5 - non-infected class 1 class 0 Actual 1 1 1 0 0 0 0 1 0 0 0 0 0 Predicted 0 0 0 1 1 1 1 0 0 0 0 0 0 <table border="1" data-bbox="411 920 596 1066"> <tr> <td></td> <th>Actual 1</th> <th>Actual 0</th> </tr> <tr> <th>Predicted 1</th> <td>1 TP</td> <td>4 FP</td> </tr> <tr> <th>Predicted 0</th> <td>3 FN</td> <td>5 TN</td> </tr> </table> Precision = $\frac{TP}{TP+FP}$ Recall = $\frac{TP}{TP+FN}$ $F_1 = \frac{2 * Precision * Recall}{Precision + Recall}$ Precision = $\frac{1}{1+4} = \frac{1}{5} = 0.2$ Recall = $\frac{1}{1+3} = \frac{1}{4} = 0.25$ $F_1 = \frac{2 * \frac{1}{5} * \frac{1}{4}}{\frac{1}{5} + \frac{1}{4}} = \frac{2 * 0.2 * 0.25}{0.2 + 0.25} = \frac{0.1}{0.45} = 0.22$		Actual 1	Actual 0	Predicted 1	1 TP	4 FP	Predicted 0	3 FN	5 TN	K3	CO1	
	Actual 1	Actual 0											
Predicted 1	1 TP	4 FP											
Predicted 0	3 FN	5 TN											
6.	Interpret the need of bias, activation function and learning rate in training a neural network. Elaborate your answer on each parameter. ⑥. Need for bias - The fixed/constant value as input to the neuron along with weight to deal with all inputs with zeros. $b = -1, w_0 = 1$ Activation Function → Firing of the neuron as response to weighted inputs. As a result Step Sign tanh, Sigmoid Learning Rate → Specified to make the network learn faster or slower. $0.1 < \eta < 0.4$; usually given: 0.1, 0.01, 0.001	K2	CO2										

7.	Outline the working of lazy learners. Illustrate the K-Nearest Neighbor Algorithm with necessary diagrams. ① lazy learners do not train the model, instead they will take test data & check how close is the test data to the training data.  ② Find K-nearest neighbors (training instances) to the test data to assign class label to test data ③ Take the majority occurrence among k-occurrences, as the class label to test data	K2	CO3	
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Part – C (2 × 12 = 24 Marks)
(Any subdivisions should be either 7+5 or 8+4)

		KL	CO	PI
8.	Consider a neural network with 2 inputs and 1 output. The output is '+1' (Plus 1) when there are even numbers of zeros or ones in the input. The output is '-1' (minus 1) when there are odd numbers of zeros or ones in the input. Model this network using a simple Perceptron Learning Algorithm and update weights when the output is misclassified. Note: Initialization values: $w_0 = w_1 = w_2 = 0.5$, Learning Rate = 0.1, bias = -1, $W_{\theta} = 1.0$. Find the updated weight of each sample for one epoch. $W_{\text{new}} = W_{\text{old}} - \eta(y_j - t_j) \cdot x_i$	K3	CO2	

$$\begin{array}{ccc}
 8. & I_1 & I_2 & 0 \\
 & 0 & 0 & 1 \\
 & 0 & 1 & 0 \\
 & 1 & 0 & 0 \\
 & 1 & 1 & 1
 \end{array}$$

$$w_0 = w_1 = w_2 = 0.5$$

$$\eta = 0.1,$$

$$b_0 = -1$$

$$w_0 = 1$$

1st Input: 0, 0 $t=1, y=0$ update weight

$$\underbrace{(-1 \times 1)}_{\text{bias, weight}} + \underbrace{(0 \times 0.5)} + \underbrace{(0 \times 0.5)} = -1 = y$$

neuron doesn't fire

$$w_{\text{new}} = w_{\text{old}} - \eta (y - t_j) x_j$$

$t=1,$

$$\begin{aligned}
 w_0 &= 1 - 0.1(-1 - 1)(-1) = 1 - 0.1(-2)(-1) \\
 &= 1 - 0.1 = 0.9
 \end{aligned}$$

update weights.

$$w_0 = 0.9$$

$$w_1 = 0.5 - 0.1(-1 - 1)0 = 0.5$$

$$w_2 = 0.5 - 0.1(-1 - 1)0 = 0.5$$

2nd Input : 0, 1, $t_1 = 0$, $y = 0$, $y = t$, Don't update weights.

$$w_0 = 0.9, w_1 = 0.5, w_2 = 0.5$$

$$(-1 \times 0.9) + (0 \times 0.5) + (1 \times 0.5) = -0.9 + 0 + 0.5 = -0.4 < 0$$

Neuron doesn't fire.

3rd Input : 1, 0, $t = 0$, $y = 0$, $y = t$, Don't update weights.

$$w_0 = 0.9, w_1 = 0.5, w_2 = 0.5$$

$$(-1 \times 0.9) + (1 \times 0.5) + (0 \times 0.5) = -0.9 + 0.5 + 0 = -0.4 < 0$$

Neuron doesn't fire.

4th Input : 1, 1, $t = 1$, $y = 1$, $y = t$, Don't update weights.

$$w_0 = 0.9, w_1 = 0.5, w_2 = 0.5$$

$$(-1 \times 0.9) + (1 \times 0.5) + (1 \times 0.5) = -0.9 + 0.5 + 0.5 = 0.1 > 0$$

Neuron will be fired.

End of Iteration 1

Bias	I_1	I_2	w_0	w_1	w_2	y	t	w_{0_new}	w_{1_new}	w_{2_new}
-1	0	0	1	0.5	0.5	0	1	0.9	0.5	0.5
-1	0	1	1	0.5	0.5	0	0	0.9	0.5	0.5
-1	1	0	1	0.5	0.5	0	0	0.9	0.5	0.5
-1	1	1	1	0.5	0.5	1	1	0.9	0.5	0.5

(Or)

9. Use the given data to compute the correlation coefficient and build a regression model for finding the weight of a person with the input as height of a person.

Height (X): 70, 62, 65, 60, 62, 72, 68, 66

Weight (Y): 215, 180, 185, 160, 162, 220, 190, 185

If the height is 63 inches, can you predict the weight of this person using the model you built?

K3

CO2

⑪ Linear Regression $\bar{y} = bx + A$

⑧ $A = M_y - b \cdot M_x$ $b = r \cdot \frac{\sigma_y}{\sigma_x}$

$$r = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^N (x_i - \bar{x})^2 \sum_{i=1}^N (y_i - \bar{y})^2}}$$

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2}$$

$M_x = \text{mean of } x = 65.625$ $\sigma_x = \sqrt{\frac{123.875}{8}}$

$M_y = \text{mean of } y = 187.125$ $\sigma_y = 20.288$

$$r = \frac{608.375}{\sqrt{123.875 \times 3292.875}} = \frac{608.375}{628.694} = 0.9525$$

$$b = r \cdot \frac{\sigma_y}{\sigma_x} = 0.9525 \times \frac{20.288}{3.935} = 4.91$$

$$A = M_y - b M_x = 187.125 - 322.270 = 187.125 - (4.91 \times 65.625) = -135.15$$

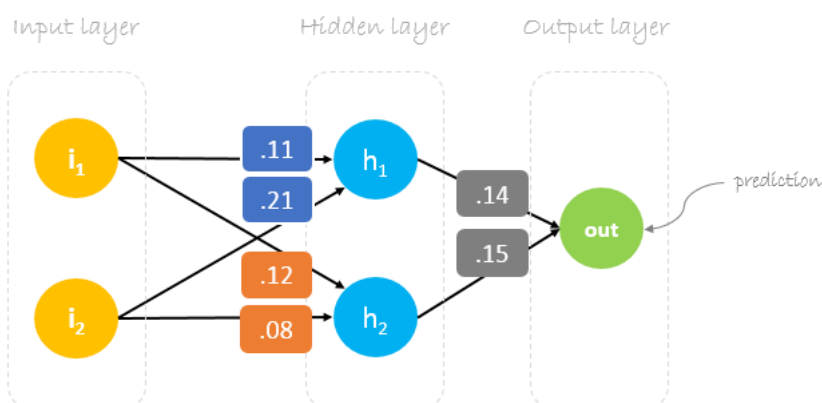
For $x = 63$, $\bar{y} = 4.91 \times 63 - 135.15$

$$= 309.33 - 135.15$$

$$\bar{y} = 174.18$$

10.

Apply Multi-Layer Perceptron algorithm for the given neural network and backpropagation the errors to reduce misclassification rate. Note: Two inputs are 2 and 3 respectively. Target output is 1. $w_1 = 0.11$, $w_2 = 0.21$, $w_3 = 0.12$, $w_4 = 0.08$, $w_5 = 0.14$ and $w_6 = 0.15$. Consider the learning rate as 1. Calculate the output of the model in the forward pass. Compute the residue and back propagate the error, update the weights. Perform one forward and one backward pass. (6 +6)

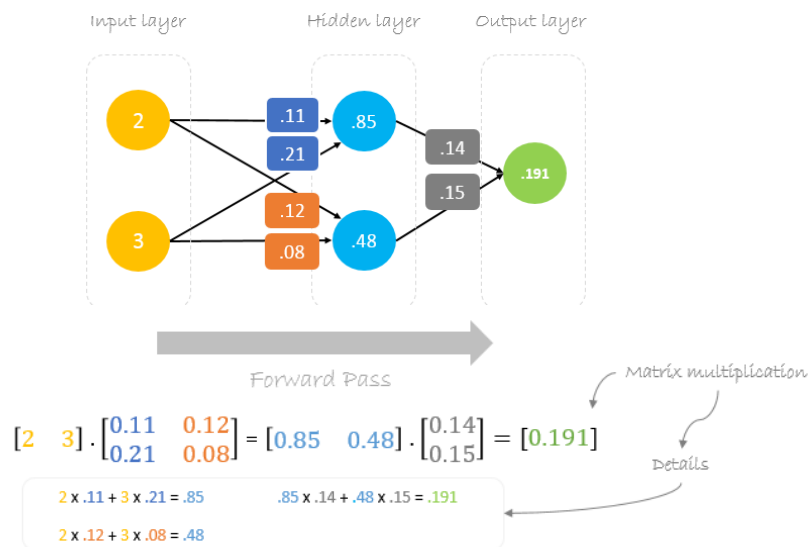


K3

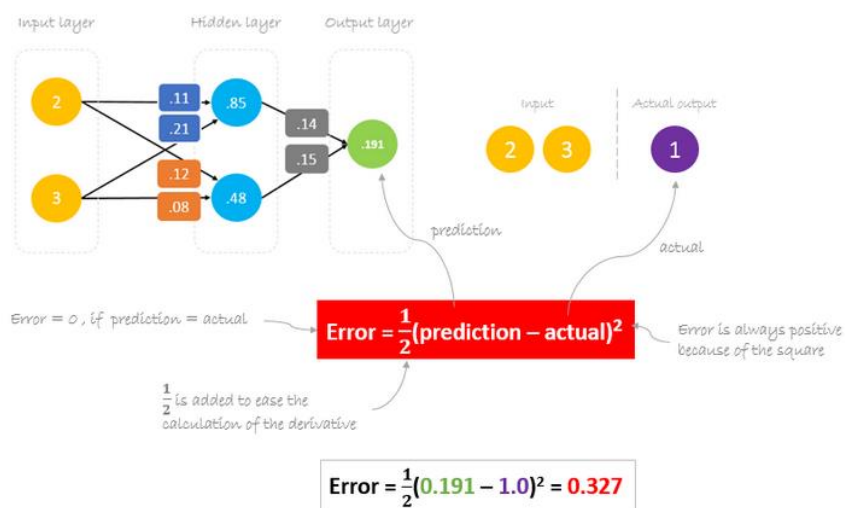
CO2

Forward Pass

We will use given weights and inputs to predict the output. Inputs are multiplied by weights; the results are then passed forward to next layer.



Calculating Error:



Backpropagation

$$*W_x = W_x - a \left(\frac{\partial \text{Error}}{\partial W_x} \right)$$

Old weight

Derivative of Error with respect to weight

New weight

Learning rate

Backward Pass

Using derived formulas we can find the new **weights**.

Learning rate: is a hyperparameter which means that we need to manually guess its value.

$$\Delta = 0.191 - 1 = -0.809$$

Delta = prediction - actual

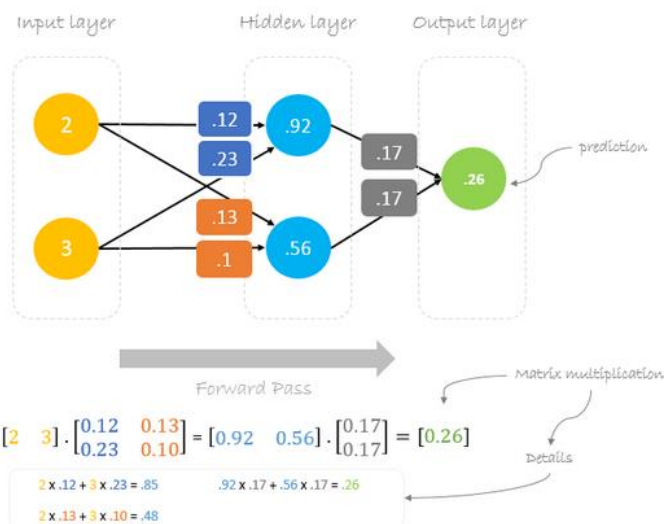
$$a = 0.05$$

Learning rate, we smartly guess this number

$$\begin{bmatrix} w_3 \\ w_6 \end{bmatrix} = \begin{bmatrix} 0.14 \\ 0.15 \end{bmatrix} - 0.05(-0.809) \begin{bmatrix} 0.85 \\ 0.48 \end{bmatrix} = \begin{bmatrix} 0.14 \\ 0.15 \end{bmatrix} - \begin{bmatrix} -0.034 \\ -0.019 \end{bmatrix} = \begin{bmatrix} 0.17 \\ 0.17 \end{bmatrix}$$

$$\begin{bmatrix} w_1 & w_3 \\ w_2 & w_4 \end{bmatrix} = \begin{bmatrix} .11 & .12 \\ .21 & .08 \end{bmatrix} - 0.05(-0.809) \begin{bmatrix} 2 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} 0.14 & 0.15 \end{bmatrix} = \begin{bmatrix} .11 & .12 \\ .21 & .08 \end{bmatrix} - \begin{bmatrix} -0.011 & -0.012 \\ -0.017 & -0.018 \end{bmatrix} = \begin{bmatrix} .12 & .13 \\ .23 & .10 \end{bmatrix}$$

Now, using the new **weights** we will repeat the forward passed



We can notice that the **prediction** 0.26 is a little bit closer to **actual output** than the previously predicted one 0.191. We can repeat the same process of backward and forward pass until **error** is close or equal to zero.

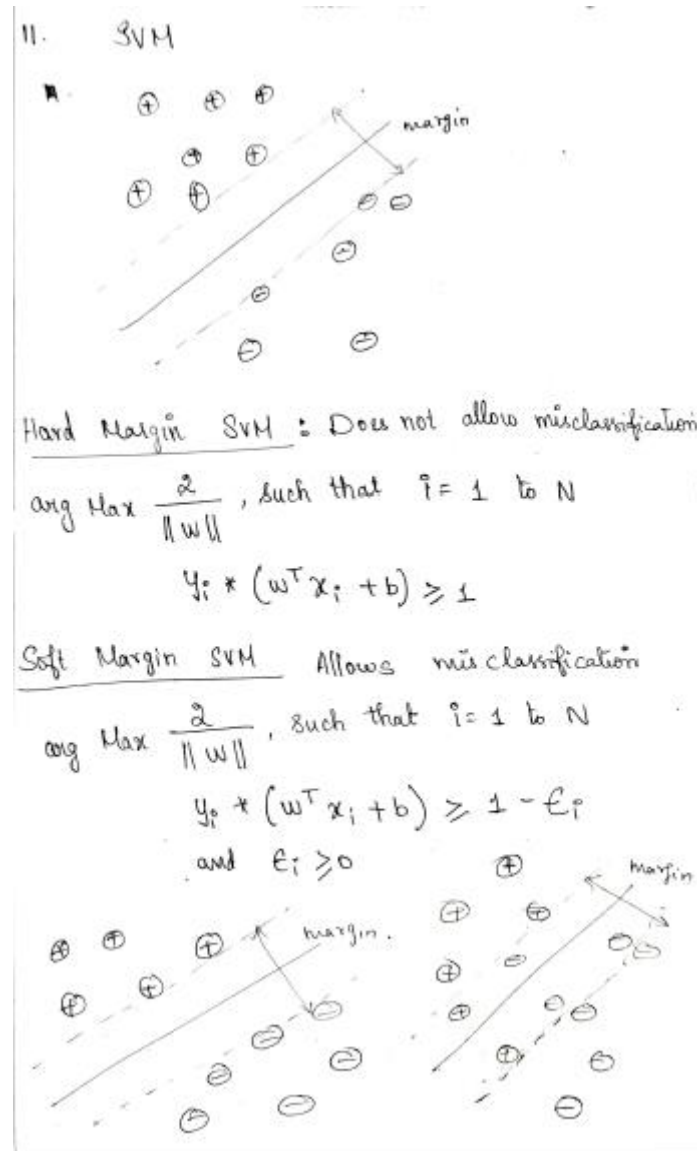
(Or)

11.	Consider a binary classification problem which separates data points into classes using a straight line. The distance between the straight line and	K3	CO2	
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the data points should be maximized. Identify the suitable classification algorithm for this specification and model using a constrained optimization problem by applying linear algebra. (5)

Derive the function for the model which tolerates and does not tolerate errors. (4)

Extend the model function to cater for non-linearly separable data. (3)



Constrained Optimization Problem.

Dual form of SVM.

$$\begin{aligned} & \max_{\alpha_i} \left[\sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j + y_i y_j + x_i^T x_j \right] \\ & \text{Such that } 0 \leq \alpha_i \leq C \\ & \sum_{i=1}^n \alpha_i y_i = 0. \end{aligned}$$

$$\max \frac{2}{\|w\|}$$

Subject to

$$1 - y_i (w^T x_i + b) \leq 0 \quad \forall i = 1, \dots, n$$

For Non-linear data

Use Kernel Trick to transform features from 2D to higher dimensional space.

Kernel Functions.

Polynomial kernel of degree 'm'

$$K(x_1, x_2) \Rightarrow (x_1^T x_2 + c)^m \quad \text{constant} = 1$$

Quadratic kernel $\equiv$ Polynomial kernel of degree 2

$$K(x_1, x_2) \Rightarrow (x_1^T x_2 + 1)^2$$

$$\text{Cubic kernel } (x_1^T x_2 + 1)^3$$

Radial Basis function

$$K(x_1, x_2) \Rightarrow e^{-\frac{\|x_1 - x_2\|^2}{2\sigma^2}}$$

Sigmoid kernel

$$K(x, x_2) = \tanh(k x_i^T y_j + \theta)^n \quad \text{on all } k \text{ and } \theta.$$

$$\max_{\alpha_i} \left[\sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j + y_i y_j + K(x_i^T x_j) \right]$$